

Colin Sauder

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Education

Colorado School of Mines

3.64 Final GPA

- Bachelors of Science in Engineering Physics

August 2024 - May 2026

4.00 GPA

- Masters of Science in Mechanical Engineering
- Graduate Certificate in Space Resources

May 2026 - May 2027

Relevant coursework:

Orbital Mechanics, Additive Manufacturing, Advanced Dynamics, Mathematical Physics, Advanced Electrodynamics, Advanced Lab

University of Denver

September 2022 - May 2024 (Transferred)

3.87 Final GPA

- Bachelors of Science in Physics (Progress Transferred to Colorado School of Mines)
- Minor in Mathematics

Relevant Coursework:

Statics, Programing (Python), Astrophysics: Radiative Process, Error and Uncertainty Analysis, Modern Physics 1 & 2, Modern Physics Lab

Projects / Coursework

Cryogenic Magnetometer

- Helped assemble and test a graphene hall sensor based magnetometer to work in tens of millikelven

Van Der Graff Generator

- Built a working, low voltage, Van Der Graff generator independently from a vinyl / nylon triboelectric effect.
- Used it to demonstrate the potential for a lint roller using electrostatic force.

Refining a Vacuum system for eventual use for DC Sputtering

- Designed a latching system to ensure proper sealing for the initial pump down of the chamber.
- Designed and plumbed the water cooling system to back the turbo and DC sputter head.
- Designed and machines the sample mount and shutter for DC sputtering
- Wrote documentation for proper operation and for fixing common issues with the chamber

Field Session

- Created a receiver for a spark gap transmitter using physical circuitry and labview to output a morse code readout using the length of the spark produced.
- Worked in a small team to create a 3d projection using both linearly and circularly polarized laser light.

Jewelrymaking

- Created several different jewelry pieces (rings, pendants, bracelets) through casting and soldering.

Work Experience

The Knight School, Denver, CO

August 2026 - Current

Chess Coach

- Taught children (6-9) how to play chess and fundamental chess strategies

Colorado School of Mines, Golden, CO

August 2025 - December 2025

Physics 2 Studio TA

- Reinforced foundational electromagnetic concepts for students during studio.
- Taught students how to build and test basic circuitry using oscilloscopes and DMM's

Bergen Bark Inn, Evergreen, CO

May 2024 - September 2025

Kennel Tech / Pet Pro

- Cleaned dog kennels and ensured the wellbeing and health of the up to 50 animals in our care.
- Worked independently and in small teams (2-4) to close the kennel.

Community Involvement

Denver Museum of Nature and Science

March 2025 - Present

Facilitator volunteer

- Helped facilitate childhood and community learning in various scientific fields using museum carts.
- Communicated and taught foundational concepts of experimentation and robotics to children through hands-on learning.

Skills

Technical

- Microsoft Excel, Python, C++, SolidWorks, Arduino, Wolfram Mathematica, HTML/CSS, Labview

Professional

- Technical Reports, Teamwork, Practical Communication, Data Collection, Data Analysis